Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/08/2005
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Interaction Of Touch And Gravitropic Signaling In Plant Roots		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	08/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1594		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Touch and gravity are two of the many stimuli that plants must integrate to generate an appropriate growth response. Due to the mechanical nature of both of these signals, shared signal transduction elements could well form the basis of the cross-talk between these two sensory systems. Two of the most widely proposed candidate signal transduction elements in both plant touch and gravity sensing are changes in Ca2+ and pH in the cytoplasm of the sensory cells. However, the precise spatial and temporal characteristics of changes in these signaling elements in response to mechanical and/or gravity stimulation, and especially how they might interact in a signaling cross-talk system, are poorly defined. The overall goal of this experimental program is therefore to seek to define the spatial and temporal character of the ionic signaling associated with mechano-perception. Using this characterization we will then mimic or block the appropriate dynamics of these ionic changes and so define how they regulate the signaling systems responsible		

	for the gravitropic response and generation of plant form. Touch stimulation has dramatic effects on plant morphogenesis and its interaction with gravimorphogenesis has a critical role in determining overall plant development and productivity. Thus, understanding the molecular mechanisms behind touch and gravity sensing systems and how these interact to generate plant form will not only contribute to our fundamental understanding of how plants grow, but may also suggest avenues for novel approaches to engineering crop stature or growth habit. In addition, it may be possible to manipulate the gravity response system of the plant with appropriate mechanical stimulation to circumvent problems of plant growth in the weightless environment of space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is employing state-of-the-art cell biological tools to elucidate the molecular and cellular mechanisms that translate the force of gravity into oriented growth of the plant. In addition, the research aims to define how other signals perceived by the cell might impact on this regulatory system and the molecular mechanisms whereby plants exhibit touch sensitivity. An improved understanding of the regulation of plant gravity sensing mechanisms and how these are altered by the plant's environment should help design approaches to enhance plant growth in microgravity, providing increased plant productivity during sustained space flight and increased control over future spaceflight experiments.
Task Progress:	We have found that touch stimulation of the root cap inhibits the cytosolic pH changes associated with gravity signaling in the sensory cells of this organ. Further, touch stimulation elicits a Ca2+ increase in the peripheral cells of the cap that propagates to the gravity sensing cells of the root cap columella. Artificially mimicking this Ca2+ change using caged-Ca2+ or caged-Ca2+ ionophore is capable of suppressing the pH changes associated with the gravitropic signaling in a similar manner to touch stimulation. Initial generation of the touch-induced Ca2+ transient is inhibited by the microtubule antagonizing drugs taxol and oryzalin suggesting that the dynamic microtubule cytoskeleton is necessary to generate the signal. However, these inhibitors do not alter gravity-related pH signaling events, implying that the touch, but not the gravity sensor is dependent on a microtubule associated signaling system. These observations suggest that touch and gravity interact in the root cap via their ionic signaling systems.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Hou G, Kramer VL, Wang YS, Chen R, Perbal G, Gilroy S, Blancaflor EB. "The promotion of gravitropism in Arabidopsis roots upon actin disruption is coupled with the extended alkalinization of the columella cytoplasm and a persistent lateral auxin gradient." Plant J. 2004 Jul;39(1):113-25. PMID: 15200646 , Jul-2004
Articles in Peer-reviewed Journals	Bibikova TN, Assmann SM, Gilroy S. "Ca2+ and pH as integrating signals in transport control." Annual Plant Reviews 2004 Nov;15:252-78. , Nov-2004